STUDYLAB HUB · AP & COLLEGE BIOLOGY · UNIT 5 (CHAPTERS 13–15)

Unit 5: Heredity, Meiosis & Genetic Analysis

Meiosis I vs. II · Crossing Over & Independent Assortment · Pedigrees · Linked Genes & Recombination · Chi-Square Test

AP & College Biology — Unit 5 (Chapters 13–15): Meiosis, Mendelian Genetics, Non-Mendelian Inheritance & Chi-Square

Independent, original study guide for AP & College Biology (informed by OpenStax open textbooks, CC BY 4.0).

1. Core Unit Roadmap

Section	Core Principle & Definitions	High-Yield Exam Application
Part 1 · Fundamental Concepts	Meiosis I vs. II	Identifying core variables, boundary conditions, and conserved quantities
Part 2 · Quantitative Models & Mechanisms	Crossing Over & Independent Assortment	Applying governing equations and predicting how system state changes under perturbation
Part 3 · Advanced Problem Solving	Chi-Square Test	Avoiding common sign, unit, and conceptual traps on multiple-choice and free-response questions

2. Essential Principles & Step-by-Step Analysis

1. **First-Principles Mechanism**:

- Every problem in **Unit 5 (Chapters 13–15): Meiosis, Mendelian Genetics, Non-Mendelian Inheritance & Chi-Square** reduces to identifying the underlying structure, forces, rates, or invariants before plugging in numbers.
- Always state the governing principle explicitly at the start of any free-response justification.

2. **High-Yield Comparison & Edge Cases**:

- Distinguish between equilibrium/steady-state behavior and transient/limiting behavior.
- Verify dimensional consistency and limiting cases (e.g., as distance or concentration approaches zero or infinity).

3. **Worked Exam Strategy**:

- Write out the known quantities with units, select the exact relation connecting them, solve symbolically first, and then substitute numerical values.